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practical 3

3.1.1

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using ndim
- The shape of the array using shape
- The total number of elements in the array using size

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, `rows` and `cols`, representing the number of rows and the number of columns.
- The next `rows` lines contain `cols` space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use `reshape()` function to reshape the input array with the specified number of rows and columns.

```
import numpy as np
```

```
rows,cols = map(int,input().split())
```

```

elements = []

for _ in range(rows) :
    elements.extend(map(int,input().split()))

array = np.array(elements).reshape(rows,cols)

print(array)

print(array.ndim)

print(array.shape)

print(array.size)

```

output :

The screenshot shows a code execution environment with the following details:

- Performance Metrics:**
 - Average time: 0.011 s (11.33 ms)
 - Maximum time: 0.022 s (22.00 ms)
- Test Results:**
 - 2 out of 2 shown test case(s) passed
 - 10 out of 10 hidden test case(s) passed
- Test Case 1 Details:**
 - Duration: 22 ms
 - Buttons: Debug, Menu, List
- Output Comparison:**

Expected output	Actual output
3 4	3 4
1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
9 10 11 12	9 10 11 12
[[1 2 3 4]]	[[1 2 3 4]]
[[5 6 7 8]]	[[5 6 7 8]]
- Interface Elements:**
 - Terminal tab (active)
 - Test cases tab

3.2.1

The given code takes two matrices, , and , as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. **Addition** ()
2. **Subtraction** ()
3. **Element-wise Multiplication** ()
4. **Matrix Multiplication** ()

5. Transpose of Matrix A

Input Format:

- The user will input 3 rows for , each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for , each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

code :

```
import numpy as np
```

```
# Input matrices
```

```
print("Enter Matrix A:")
```

```
matrix_a = np.array([list(map(int, input().split())) for i in range(3)])
```

```
print("Enter Matrix B:")
```

```
matrix_b = np.array([list(map(int, input().split())) for i in range(3)])
```

```
# Addition
```

```
addition_result = matrix_a + matrix_b
```

```
print("Addition (A + B):")
print(addition_result)

# Subtraction
sub_result = matrix_a - matrix_b
print("Subtraction (A - B):")
print(sub_result)

# Multiplication (element-wise)
elemental_multi_result = matrix_a * matrix_b
print("Element-wise Multiplication (A * B):")
print(elemental_multi_result)

# Matrix multiplication (dot product)
matrix_dotproduct_result = np.dot(matrix_a , matrix_b )
print("A dot B:")
print(matrix_dotproduct_result)

# Transpose
transpose_a = matrix_a.T
print("Transpose of A:")
print(transpose_a)
```

output :

3.2.2

You are given two arrays `arr1` and `arr2`. You need to perform horizontal and vertical stacking operations on them using NumPy.

- **Horizontal Stacking:** Stack the two matrices horizontally (side by side).
- **Vertical Stacking:** Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Code :

```
import numpy as np
```

```
# Input matrices
```

```

print("Enter Array1:")

arr1 = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Array2:")

arr2 = np.array([list(map(int, input().split())) for i in range(3)])

# Perform horizontal stacking (hstack)

h_stack = np.hstack((arr1, arr2))

print("Horizontal Stack:")

print(h_stack)

```

```

# Perform vertical stacking (vstack)

v_stack = np.vstack((arr1, arr2))

print("Vertical Stack:")

print(v_stack)

```

output :

The screenshot shows a code editor with the following code:

```

17 # Perform vertical stacking (vstack)
18 v_stack = np.vstack((arr1, arr2))

```

Below the code, the execution results are displayed:

- Average time:** 0.024 s (24.25 ms)
- Maximum time:** 0.036 s (36.00 ms)
- Test Results:**
 - 2 out of 2 shown test case(s) passed
 - 2 out of 2 hidden test case(s) passed

Test case 1 (36 ms):

Expected output	Actual output
Enter-Array1: 1 2 3	Enter-Array1: 1 2 3
4 5 6	4 5 6
7 8 9	7 8 9
Enter-Array2: 4 5 6	Enter-Array2: 4 5 6

At the bottom, there are tabs for "Terminal" and "Test cases", and a navigation bar with buttons: "< Prev", "Reset", "Submit", and "Next >".

3.2.3

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

Code:

```
import numpy as np
```

```
# Take user input for the start, stop, and step of the sequence
```

```
start = int(input())
```

```
stop = int(input())
```

```
step = int(input())
```

```
# Generate the sequence using np.arange()
```

```
sequence = np.arange(start, stop, step)
```

```
# Print the generated sequence
```

```
print(sequence)
```

output:

The screenshot shows a test results interface. At the top, there are two boxes for timing: 'Average time' with '0.014 s' and '14.25 ms', and 'Maximum time' with '0.026 s' and '26.00 ms'. To the right, two green status boxes indicate '2 out of 2 shown test case(s) passed' and '2 out of 2 hidden test case(s) passed'. Below this, 'Test case 1' is expanded, showing a '26 ms' execution time and a 'Debug' button. It contains a table comparing 'Expected output' and 'Actual output'. The expected output is a list [3, 10, 2] and the array [3, 5, 7, 9]. The actual output is identical. 'Test case 2' is partially visible below.

Expected output	Actual output
3	3
10	10
2	2
[3, 5, 7, 9]	[3, 5, 7, 9]

3.2.4

You are given two arrays A and B. Your task is to complete the function `array_operations`, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:

- Compute the element-wise sum, difference, and product of the two arrays.

2. Statistical Operations:

- Calculate the mean, median, and standard deviation of array A.

3. Bitwise Operations:

- Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: $A_i \text{ OR } B_i$).

Input Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

Code :

```
import numpy as np
```

```
def array_operations(A, B):
```

```
    # Convert A and B to NumPy arrays
```

```
    A = np.array(A)
```

```
    B = np.array(B)
```

```
    # Arithmetic Operations
```

```
    sum_result = A + B
```

```
    diff_result = A - B
```

```
    prod_result = A * B
```

```
    # Statistical Operations
```

```
    mean_A = np.mean(A)
```

```
    median_A = np.median(A)
```

```
    std_dev_A = np.std(A)
```

```
    # Bitwise Operations
```

```
    and_result = A & B
```

```
    or_result = A | B
```

```
    xor_result = A ^ B
```

```
    # Output results with one space between each element
```

```
    print("Element-wise Sum:", ' '.join(map(str, sum_result)))
```

```
    print("Element-wise Difference:", ' '.join(map(str, diff_result)))
```

```
    print("Element-wise Product:", ' '.join(map(str, prod_result)))
```

```

print(f"Mean of A: {mean_A}")
print(f"Median of A: {median_A}")
print(f"Standard Deviation of A: {std_dev_A}")

print("Bitwise AND:", ' '.join(map(str, and_result)))
print("Bitwise OR:", ' '.join(map(str, or_result)))
print("Bitwise XOR:", ' '.join(map(str, xor_result)))

```

```

A = list(map(int, input().split())) # Elements of array A
B = list(map(int, input().split())) # Elements of array B
array_operations(A, B)

```

output:

The screenshot shows a code execution interface with the following components:

- Code Editor:** Lines 26 and 27 of code are visible:


```

26 >print("Element-wise Sum:", ' '.join(map(str, sum_result)))
27 >print("Element-wise Difference:", ' '.join(map(str, diff_result)))
      
```
- Performance Metrics:**
 - Average time: 0.029 s (29.00 ms)
 - Maximum time: 0.071 s (71.00 ms)
- Test Results:**
 - 1 out of 1 shown test case(s) passed
 - 2 out of 2 hidden test case(s) passed
- Test Case 1 (71 ms):**
 - Expected output:**

```

1 2 3 4
5 6 7 8
Element-wise Sum: -6 -8 -10 -12
Element-wise Difference: -4 -4 -4 -4
Element-wise Product: -5 -12 -21 -32
Mean of A: -2.5
      
```
 - Actual output:**

```

1 2 3 4
5 6 7 8
Element-wise Sum: -6 -8 -10 -12
Element-wise Difference: -4 -4 -4 -4
Element-wise Product: -5 -12 -21 -32
Mean of A: -2.5
      
```
- Interface:** Includes a 'Terminal' tab and a 'Test cases' button.

3.2.5

he given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the original_array and assigning it to view_array.
- Creating a copy of the original_array and assigning it to copy_array.

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

Original array after modifying view: <original_array>

View array: <view_array>

- After modifying the copy:

Original array after modifying copy: <original_array>

Copy array: <copy_array>

Code and output :

```
1 import numpy as np
2
3 inputlist = list(map(int,input().split(" ")))
4
5 # Original array
6 original_array = np.array(inputlist)
7
8 # Create a view
9 view_array = original_array.view()
10
11 # Create a copy
12 copy_array = original_array.copy()
13
14 # Modify the view
15 view_array[0] = 99
16 print("Original array after modifying view:", original_array)
17 print("View array:", view_array)
```

Average time 0.009 s 9.25 ms Maximum time 0.016 s 16.00 ms 2 out of 2 shown test case(s) passed 2 out of 2 hidden test case(s) passed

✓ Test case 1 16 ms

✓ Test case 2 6 ms

3.2.6

The given code in the editor takes a single array, array1, as space-separated integers as input from the user.

Additionally, it takes the following inputs:

- search_value: The value to search for in the array.

- `count_value`: The value to count its occurrences in the array.
- `broadcast_value`: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

- 1. Searching:** Find the indices where `search_value` appears in `array1` and print these indices.
- 2. Counting:** Count how many times `count_value` appears in `array1` and print the count.
- 3. Broadcasting:** Add `broadcast_value` to each element of `array1` using broadcasting, and print the resulting array.
- 4. Sorting:** Sort `array1` in ascending order and print the sorted array.

Input Format:

1. A single line containing space-separated integers representing `array1`.
2. An integer `search_value` represents the value to search for in the array.
3. An integer `count_value` represents the value to count in the array.
4. An integer `broadcast_value` represents the value to add to each element of the array.

Output Format:

1. The indices where `search_value` occurs in `array1`.
2. The count of occurrences of `count_value` in `array1`.
3. The array after adding the `broadcast_value` to each element.
4. The sorted array.

code and output :

```
arrayOpe...
1 import numpy as np
2
3 # Input array from the user
4 array1 = np.array(list(map(int, input().split())))
5
6 # Searching
7 search_value = int(input("Value to search: "))
8 count_value = int(input("Value to count: "))
9 broadcast_value = int(input("Value to add: "))
10
11 # Find indices where value matches in array1
12 a=np.where(array1==search_value)[0]
13 print(a)
14 # Count occurrences in array1
15 b=np.count_nonzero(array1==count_value)
16 print(b)
17
18 # Broadcasting addition
19 c=array1 + broadcast_value
20 print(c)
21
22 # Sort the first array
23 d=np.sort(array1)
24 print(d)
```

Average time 0.026 s 26.00 ms Maximum time 0.057 s 57.00 ms 2 out of 2 shown test case(s) passed 2 out of 2 hidden test case(s) passed

Test case 1 Expected output Actual output

4 1 1 2 2 2	4 1 1 2 2 2
Value to search: 1	Value to search: 1
Value to count: 2	Value to count: 2
Value to add: 2	Value to add: 2
[0 1 2]	[0 1 2]
3	3

Terminal Test cases

3.2.7

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details:** Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students:** Determine the total number of students in the dataset.
- **Print all student roll numbers:** Extract and print the roll numbers of all students.
- **Print Subject 1 marks:** Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2:** Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3:** Identify the highest marks in Subject 3.
- **Print all subject marks:** Display the marks of all students for each subject.
- **Find total marks of students:** Compute the total marks for each student across all subjects.
- **Find the average marks of each student:** Compute the average marks for each student.
- **Find average marks of each subject:** Compute the average marks for all students in each subject.

- **Find average marks of Subject 1 and Subject 2:** Compute the average marks for Subject 1 and Subject 2.
- **Find average marks of Subject 1 and Subject 3:** Compute the average marks for Subject 1 and Subject 3.
- **Find the roll number of the student with maximum marks in Subject 3:** Identify the student with the highest marks in Subject 3 and print their roll number.
- **Find the roll number of the student with minimum marks in Subject 2:** Identify the student with the lowest marks in Subject 2 and print their roll number.
- **Find the roll number of students who scored 24 marks in Subject 2:** Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- **Find the count of students who got less than 40 marks in Subject 1:** Count the number of students who scored less than 40 marks in Subject 1.
- **Find the count of students who got more than 90 marks in Subject 2:** Count the number of students who scored more than 90 marks in Subject 2.
- **Find the count of students who scored ≥ 90 in each subject:** Count the number of students who scored 90 or more marks in each subject.
- **Find the count of subjects in which each student scored ≥ 90 :** Determine how many subjects each student scored 90 or more marks in.
- **Print Subject 1 marks in ascending order:** Sort and print the marks of students in Subject 1 in ascending order.
- **Print students who scored between 50 and 90 in Subject 1:** Display students who scored marks between 50 and 90 in Subject 1.
- **Find index positions of students who scored 79 in Subject 1:** Identify the index positions of students who scored exactly 79 marks in Subject 1.

Note: Fill in the missing code to perform the above-mentioned operations.

Code :

```
import numpy as np
```

```
a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)
```

```
# 1. Print all student details
```

```
print("All student Details:\n", a )
```

2. print total students

```
print("Total Students:",a.shape[0] )
```

3. Print all student Roll numbers

```
print("All Student Roll Nos", a[:,0] )
```

4. Print subject 1 marks

```
print("Subject 1 Marks", a[:,1])
```

5. print minimum marks of Subject 2

```
print("Min marks in Subject 2", np.min(a[:,2] ))
```

6. print maximum marks of Subject 3

```
print("Max marks in Subject 3", np.max(a[:,3]) )
```

7. Print All subject marks

```
print("All subject marks:", a[:,1:] )
```

8. print Total marks of students

```
print("Total Marks", np.sum(a[:,1:],axis=1) )
```

9. print average marks of each student

```
avg=np.mean(a[:,1:],axis=1)
```

```
print(np.round(avg,1))
```

10. print average marks of each subject

```
print("Average Marks of each subject", np.mean(a[:,1:],axis=0) )
```

```
# 11. print average marks of S1 and S2
```

```
print("Average Marks of S1 and S2", np.mean(a[:,1:3],axis=0 ) )
```

```
# 12. print average marks of S1 and S3
```

```
print("Average Marks of S1 and S3", np.mean(a[:,[1,3]],axis=0) )
```

```
# 13. print Roll number who got maximum marks in Subject 3
```

```
i=np.argmax(a[:,3])
```

```
print("Roll no who got maximum marks in Subject 3", a[i,0] )
```

```
# 14. print Roll number who got minimum marks in Subject 2
```

```
j=np.argmin(a[:,2])
```

```
print("Roll no who got minimum marks in Subject 2", a[j,0] )
```

```
# 15. print Roll number who got 24 marks in Subject 2
```

```
print(f"Roll no who got 24 marks in Subject 2 [{a[a[:,2]==24,0}]" )
```

```
# 16. print count of students who got marks in Subject 1 < 40
```

```
print("Count of students who got marks in Subject 1 < 40", np.sum(a[:,1]<40) )
```

```
# 17. print count of students who got marks in Subject 2 > 90
```

```
print("Count of students who got marks in Subject 2 > 90:", np.sum(a[:,2]>90) )
```

```
Count=[len(np.argwhere(a[:,1]>=90)),len(np.argwhere(a[:,2]>=90)),len(np.argwhere(a[:,3]>=90))]
```

```
count=np.array(Count)
```

18. print count of students in each subject who got marks >= 90

```
print("Count of students in each subject who got marks >= 90:", count      )
```

19. print count of subjects in which each student got marks >= 90

```
print("Roll no:", a[:,0]      )
```

```
print("Count of subjects in which student got marks >= 90:", np.sum(a[:,1:]>=90,axis=1)      )
```

20. Print S1 marks in ascending order

```
print(np.sort(a[:,1]))
```

```
print(a[(a[:,1]>=50) & (a[:,1]<=90)])
```

```
print(a[(a[:,1]>=45) & (a[:,1]<=90)])
```

```
k=np.where(a[:,1]==79)
```

```
print(k)
```

output :

```
12 print(" All Student Roll nos ",a[:,0])
13
```

Average time
0.074 s
74.00 ms

Maximum time
0.074 s
74.00 ms

1 out of 1 shown test case(s) passed

Test case 1 74 ms Debug

Expected output

All student Details:

[[301.--67.--77.--88.]

[302.--78.--88.--77.]

[303.--45.--56.--89.]

[304.--88.--98.--45.]

[305.--78.--88.--99.]

[306.--68.--78.--88.]

Actual output

All student Details:

[[301.--67.--77.--88.]

[302.--78.--88.--77.]

[303.--45.--56.--89.]

[304.--88.--98.--45.]

[305.--78.--88.--99.]

[306.--68.--78.--88.]

Terminal Test cases